Kanghwi (Kenny) Lee

Speech & Audio ML Researcher
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Summary

Speech and audio ML researcher and Ph.D. candidate at ETH Zurich (graduating July 2027), working across speech language models, audio representations and large-scale audio data. At NAVER Cloud, developed Redwing, a watermark for speech LMs that stays detectable after eight neural-codec resynthesis passes, where token-level baselines fall below 10%. Ph.D. work includes fine-tuning audio foundation models, conditional generative modeling, and a pipeline that turned 80,000 hours of recordings into about 100M clustered vocalizations with about 200 hours of human work.

Experience

NAVER Cloud, Seongnam, Korea Jul 2026–Jan 2027
Research Intern, Voice Synthesis Team

- Benchmarked public audio-watermarking methods and fine-tuned an established approach to meet the team's requirements for its speech services.
- Conducted research on robustness to neural-codec resynthesis and developed **Redwing**. Modeled codec token substitutions as a graph and used constrained optimization to jointly design watermark embedding and detection functions. Applied the watermark through speech-LM logit biasing, without retraining the model or codec.
- On Moshi, achieved **80.7% detection after eight Mimi resynthesis passes**, versus under 10% for token-level baselines, at a calibrated 1% false-positive rate.
- Redwing also generalized to TTS: **84.7% and 99.8% detection** on CosyVoice3 and MOSS-TTS after eight native codec resynthesis passes, versus under 10% for the standard token-level baseline, at comparable speech-quality cost.

ETH Zurich, Institute of Neuroinformatics, Zurich Apr 2022–Present
Ph.D. Candidate, Audio ML & Computational Neuroscience; advisor: Richard H. R. Hahnloser

- Developed an age-conditioned generative model, trained with optimal-transport pairing on about 680K vocalizations, that generates counterfactuals of each vocalization at other ages. Introduced Trajectory Variance, the spread of these predicted shifts in latent space, as a label-free score of developmental change; it separated learned song from innate calls where nonparametric baselines did not. **Interspeech 2026 oral (sole author).**
- Developed an embedding-based clustering workflow that focuses annotation on cluster definition and outlier review, so **human effort scales primarily with acoustic diversity rather than dataset size**. On a 1,454-vocalization benchmark, its clusters matched the experts' nearly as well as the experts matched each other (ARI 0.73 vs. 0.78). At scale, **about 100M vocalizations** segmented by WhisperSeg from **80,000 hours** of recordings were clustered with **about 200 hours of human work**.
- Fine-tuned **Whisper** models for audio segmentation. Second author of **WhisperSeg (ICASSP 2024)**, contributing dataset processing, baseline evaluation, and cross-domain analysis of human and animal recordings.

Technical Skills

- **Tools & Systems:** Python, PyTorch, NumPy, SciPy, Linux, Git, sharded multi-GPU training (HPC).
- **Modeling & Optimization:** Transformers, VAEs, representation learning, fine-tuning, optimal transport, spectral graph methods, constrained optimization, clustering (HDBSCAN, k-NN, UMAP).
- **Speech & Audio:** Audio segmentation/VAD, neural audio codecs (Mimi, EnCodec, SNAC, DAC), speech watermarking; speech-quality evaluation (UTMOSv2, DNSMOS Pro, NISQA).
- **Languages:** Korean (native), English (fluent; former English–Korean technical interpreter, ROK Army).

Selected Publications

- **K. Lee**, K. Jeong, J. Liu. "Tokens Change, Structure Endures: Spectral Watermarking for Generated Speech." arXiv:2609.33774, 2026. Under review at ICLR 2027. [paper, demo]
- **K. Lee**. "Trajectory Variance: An Unsupervised Measure of Developmental Vocal Plasticity in Birdsong." Interspeech 2026 (oral). [paper, code]
- N. Gu, **K. Lee**, et al. "Positive Transfer of the Whisper Speech Transformer to Human and Animal Voice Activity Detection." ICASSP 2024. [paper, code]
- **K. Lee**, M. Basha, A. T. Zai, R. H. R. Hahnloser. "Benchmarking Automated and Semi-Automated Vocal Clustering Methods." Applied Sciences, 2026. [paper, code]

Education

ETH Zurich, Ph.D. Candidate, Institute of Neuroinformatics 2022–Jul 2027 (expected)
University of Zurich, M.Sc. Neural Systems and Computation, GPA 5.5/6.0 2020–2022
Korea University, B.S. Computer Science; B.Econ. Statistics, GPA 4.02/4.50 2014–2020